```
> restart;#PSTc
```

```
> ρ := 9.9·10^-37; TQ := 1.765·10^-19; SQ := 4.25·10^-19; c := 299792458; Rs :=
    1.553479855 × 10^26; R := 4.235518529 × 10^22;
    # ρ  density of void space;
```

$$\rho := 9.900000000 \times 10^{-37}$$

$$TQ := 1.765000000 \times 10^{-19}$$

$$SQ := 4.250000000 \times 10^{-19}$$

$$c := 299792458$$

$$Rs := 1.553479855 \times 10^{26}$$

$$R := 4.235518529 \times 10^{22} \quad (1)$$

```
> V := (χ^0.5)/(ρ^0.5);
```

$$V := 1.005037815 \times 10^{18} \sqrt{\chi} \quad (2)$$

```
>  χ := 0.09070366748; V;
```

$$\chi := 0.09070366748$$

$$3.026877369 \times 10^{17} \quad (3)$$

```
> (4·Pi/3)(Rs)^3/SQ^3;# approx. number of space quantum bloks in LU
```

$$2.045684699 \times 10^{134} \quad (4)$$

```
> T = 2·Rs/V; PTU := 3·24·3600;# 3 days is the unit
```

$$T = 1.026457081 \times 10^{9}$$

$$PTU := 259200 \quad (5)$$

```
> PTU := 3·24·3600; 2·R/V;#The PTU is 3 days is the unit;
```

$$PTU := 259200$$

$$279860.5964 \quad (6)$$

```
> PTQ := evalf(SQ/c);
```

$$PTQ := 1.417647405 \times 10^{-27} \quad (7)$$

```
> V := 2·R/(3·24·3600); PTU := 3·24·3600; 2·R/V;# 3 days is the unit
```

$$V := 3.268147014 \times 10^{17}$$

$$PTU := 259200$$

$$259200.0000 \tag{8}$$

> $Vp := \frac{2 \cdot R}{PTU}$# The speed limit of a wave

$$Vp := 3.268147014 \times 10^{17} \tag{9}$$

> $\frac{Vp}{24 \cdot 3 \cdot c}$;# The speed limit of un object with null rest mass~14 $\cdot 10^6$ times the speed of light $14 \cdot 10^6$ LS

$$1.514078475 \times 10^{7} \tag{10}$$

> $\chi := \left(Vp \cdot \rho^{0.5}\right)^2$;

$$\chi := 0.1057397706 \tag{11}$$

> $4 \cdot PTQ$;

$$5.670589620 \times 10^{-27} \tag{12}$$

> $\frac{1}{R}$;# is the rough aproximation of photon radius in [m]

$$2.360986012 \times 10^{-23} \tag{13}$$

> $rb := 5.292410932 \times 10^{-11}$; rb^2;

$$rb := 5.292410932 \times 10^{-11}$$

$$2.800961347 \times 10^{-21} \tag{14}$$

> $Sq := 4.3 \cdot 10^{-19}$; Sq^2;

$$Sq := 4.300000000 \times 10^{-19}$$

$$1.849000000 \times 10^{-37} \tag{15}$$

> SQ^2;# The Quark dimmension 4.25-4.3·10^{-19}

$$1.806250000 \times 10^{-37} \tag{16}$$

> PTQ^2

$$2.009724165 \times 10^{-54} \tag{17}$$

> $F\tau := -\frac{9.897187240 \times 10^{23}}{r^2} + \frac{3.121665052 \times 10^{21}}{r} - 0.07957747155$;

$$F\tau := -\frac{9.897187240 \times 10^{23}}{r^2} + \frac{3.121665052 \times 10^{21}}{r} - 0.07957747155 \tag{18}$$

> simplify(-.9914018010e24/r^2+.3121665052e22/r-.7957747152e-1);

$$-\frac{9.914018010 \times 10^{23}}{r^2} + \frac{3.121665052 \times 10^{21}}{r} - 0.07957747152 \tag{19}$$

```
>
> restart;# The Gamma Ray spherical simetry;
> de := diff(g(r), r$2) = r·g(r)/Rs;
```

$$de := \frac{d^2}{dr^2}\, g(r) = \frac{r\, g(r)}{Rs} \tag{20}$$

```
> ics := g( 2·Rs) = 0 , D(g)(0) = R;
```

$$ics := g(2\,Rs) = 0,\ D(g)(0) = R \tag{21}$$

```
> sol := dsolve({de, ics});
```

$$sol := g(r) = -\left(2\,R\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right) - \mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) + 3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\right)\right) \tag{22}$$

```
> g1(r) := rhs(sol);
```

$$g1 := r \mapsto rhs(sol) \tag{23}$$

```
> g1(0);
```

$$-\left(2\,R\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right) - \mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3} r\right)\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right) + 3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3} Rs\right)\right)\right) \tag{24}$$

```
> g(r) := (2 R π AiryBi(-2 (-1/Rs)^(1/3) Rs) AiryAi(-(-1/Rs)^(1/3) r)) /
    (Γ(2/3) (-1/Rs)^(1/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs) + 3^(1/6) AiryBi(
    -2 (-1/Rs)^(1/3) Rs))) - (2 AiryAi(-2 (-1/Rs)^(1/3) Rs) R π AiryBi(
    -(-1/Rs)^(1/3) r)) / (Γ(2/3) (-1/Rs)^(1/3) (3^(2/3) AiryAi(-2 (-1/Rs)^(1/3) Rs)
    + 3^(1/6) AiryBi(-2 (-1/Rs)^(1/3) Rs)));
```

$$g := r \mapsto \left(2\cdot R\cdot\pi\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right) - \left(2\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot R\cdot\pi\cdot\mathrm{AiryBi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}\cdot r\right)\right) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\right) \tag{25}$$

```
> # g(r)·∂u(t)/∂t + u(t)· ∂g(r)/∂r·∂r/∂t = c ;
    # thus for r = 0 and t≠0 we have u(t) adimensional;
```

```
> eq1 := (g(0)·diff(u(t), t) + u(t)·R·V − V = 0);
```

$$eq1 := \left(\left(2\,R\,\pi\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right) \Big/ \left(3\,\Gamma\left(\frac{2}{3}\right)^{2}\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)\right)\right)\left(\frac{\mathrm{d}}{\mathrm{d}t}\,u(t)\right)+u(t)\,R\,V-V=0 \tag{26}$$

```
> dsolve(eq1, u(t));
```

$$u(t) = \frac{1}{R} + \mathrm{e}^{-\frac{3\,V\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t\,\Gamma\left(\frac{2}{3}\right)^{2}}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}c_1 \tag{27}$$

```
> u(t) := e^(-3*V*Γ(2/3)^2*(-1/Rs)^(1/3)*(3^(2/3)*AiryAi(-2*(-1/Rs)^(1/3)*Rs)+3^(1/6)*AiryBi(-2*(-1/Rs)^(1/3)*Rs))*t/(2*π*(AiryAi(-2*(-1/Rs)^(1/3)*Rs)*3^(5/6)-AiryBi(-2*(-1/Rs)^(1/3)*Rs)*3^(1/3))))/R;
```

$$+ _C1 \Bigg)$$

$$e^{-\frac{3\,V\,\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}$$

$$u := t \mapsto \left(\frac{e^{-\frac{3\cdot V\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\cdot\Gamma\left(\frac{2}{3}\right)^2\cdot t}{2\cdot\pi\cdot\left(\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{5/6}-\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/3}\right)}}}{R} \right. \tag{28}$$

$$\left. + c_1 \right) \cdot e^{-\frac{3\cdot V\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot\left(3^{2/3}\cdot\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)+3^{1/6}\cdot\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\right)\cdot\Gamma\left(\frac{2}{3}\right)^2\cdot t}{2\cdot\pi\cdot\left(-\mathrm{AiryAi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{5/6}+\mathrm{AiryBi}\left(-2\cdot\left(-\frac{1}{Rs}\right)^{1/3}\cdot Rs\right)\cdot 3^{1/3}\right)}}$$

> `sol1 := u(t)·g(r);  # H(r, t) = u(t)·g(r)`

$$sol1 := \left(\frac{e^{-\frac{3\,V\Gamma\left(\frac{2}{3}\right)^2\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t}{2\,\pi\left(\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}-\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}}{R} \right. \tag{29}$$

$$\left. + c_1 \right)$$

$$e^{-\frac{3\,V\left(-\frac{1}{Rs}\right)^{1/3}\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)t\,\Gamma\left(\frac{2}{3}\right)^2}{2\,\pi\left(-\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{5/6}+\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)3^{1/3}\right)}}$$

$$\left(\left(2\,R\,\pi \right. \right.$$

$$\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\mathrm{AiryAi}\left(-\left(-\frac{1}{Rs}\right)^{1/3}r\right)\Big) \Big/ \left(\Gamma\left(\frac{2}{3}\right)\left(-\frac{1}{Rs}\right)^{1/}\right.$$

$$\left(3^{2/3}\,\mathrm{AiryAi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)+3^{1/6}\,\mathrm{AiryBi}\left(-2\left(-\frac{1}{Rs}\right)^{1/3}Rs\right)\right)$$

$$\left.\left.\left.

